

Independent Domination in Outerplanar Graphs

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Abstract

In this paper we consider the maximum value of the independent domination number of outerplanar graphs. If one considers all outerplanar graphs or restricts to all maximal outerplanar graphs, this maximum is easy to find and known. So we focus on some subfamilies. Our main contribution is that, if one restricts to 2-connected outerplanar graphs of order n , then the maximum is $(2n + 1)/5$, and there exists an infinite family of graphs that achieve this bound whenever $2n + 1$ is a multiple of 5.

1 Introduction

In this paper we continue the study of independent domination in the family of outerplanar graphs. Recall that a dominating set S of a graph G is a set such that every vertex not in S has a neighbor in S , and it is an independent dominating set (abbreviated ID-set) if S itself is independent. The domination number $\gamma(G)$ and the independent domination number $i(G)$ are the minimum size of a dominating set and independent dominating set respectively. Note too that an independent dominating set is equivalent to a maximal independent set.

Recall that an outerplanar graph is a planar graph that can be drawn in the plane with all vertices on the outer face. A maximal outerplanar graph is an outerplanar graph where no edge can be added and the graph remains outerplanar. The basics of independent domination in outerplanar graphs are given in multiple places (including for example [6]). For example, it is well-known that an outerplanar graph is 3-colorable. Indeed, a maximal outerplanar graph has a unique 3-coloring, and for this coloring every color class is an independent dominating set. This was, for example, exploited by Fisk [4] in the simple proof

of Chvátal's art-gallery theorem, and immediately gives that $i(G) \leq n/3$ for a maximal outerplanar graph G of order n (and this is best possible). For more on the independent domination number in general, see [5]. There are also results on the domination number of maximal outerplanar graphs such as the work by Campos and Wakabayashi [1], Liu [8], Plummer et al. [10], and Tokunaga [11]; while, for example, the total domination number of maximal outerplanar graphs is investigated by Dorfling et al. [2, 3].

As we note in the next section, it is easy to determine the maximum value of the independent domination number for all outerplanar graphs or for all maximal outerplanar graphs. Thus we focus here on subfamilies. The main contribution is the following result:

Theorem 1 *If G is a 2-connected outerplanar graph of order $n \geq 5$, then $i(G) \leq (2n + 1)/5$, and this is sharp.*

The proof of Theorem 1 is given in Section 3. We provide some preliminary thoughts in Section 2, including a result on acyclic domination; and we consider some other subfamilies in Section 4.

2 General and Maximal Outerplanar Graphs

We noted above that the independent domination number of a maximal outerplanar graph is at most one-third the order. We observe next that the result of Campos and Wakabayashi [1] and Tokunaga [11] extends to independent domination. Namely, if G is a maximal outerplanar graph with n vertices, s of which have degree 2, then $i(G) \leq (n + s)/4$. The proof given by Tokunaga immediately applies. Namely, such a G has a proper 4-coloring such that every vertex of degree more than 2 has all four colors in its closed neighborhood; and thus starting with the four color classes and adding the degree-2 vertices once if needed, one obtains four independent dominating sets whose total size is at most $n + s$.

In contrast, for general outerplanar graphs the maxima for ordinary domination and independent domination are different. For ordinary domination in graphs without isolates the maximum is $n/2$. On the other hand, MacGillivray

and Seyffarth [9] provided an upper bound for the independent domination number in terms of the chromatic number of a graph. It follows from this that an outerplanar graph G of order n without isolates has $i(G) \leq 2n/3 - 1$ (provided $n \geq 3$). And this bound is sharp because of the corona K_3 with feet, illustrated in Figure 1.

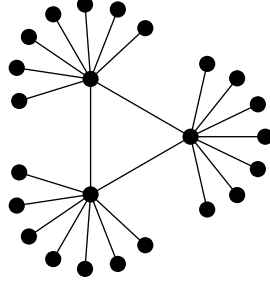


Figure 1: An outerplanar graph with independent domination about $\frac{2}{3}$ its order

In contrast, we note that half the order is an upper bound for the acyclic domination number of an isolate-free outerplanar graph. Recall that Hedetniemi et al. [7] defined the *acyclic domination number* $\gamma_a(G)$ of a graph G as the minimum cardinality of a dominating set that induces a forest. This parameter is sandwiched between the ordinary and independent domination numbers.

Observation 2 *If G is an outerplanar graph of order n without isolates, then $\gamma_a(G) \leq n/2$.*

PROOF. It is well known (and easily proved by induction) that the vertices of an outerplanar graph can be colored with red and blue such that each class induces a forest. (In other words, the arboricity is at most 2.) If one takes such a coloring that minimizes the number of vertices that have the same color as all their neighbors, then one readily obtains such a coloring where both color classes are dominating sets. The bound follows. QED

3 Proof of the Main Theorem

3.1 Overview and Reductions

The proof of the bound in Theorem 1 is by induction, both on the order of G and then on the number of edges of G . The base case is a cycle where the result is true (since $i(C_n) = \lceil n/3 \rceil$). Note that it is easy to show that a 2-connected outerplanar graph is hamiltonian; we refer to the hamilton cycle as the *outer cycle* and refer to the other edges as *chords*. Further, we define a *portion* of the outer cycle as the vertices of some subpath of that cycle.

The proof can be thought of as in two halves. We define a chord as *short* if it joins two vertices that are two or three apart on the outer cycle; that is, there is either one or two vertices on the outer cycle between the endpoints of the chord. The first half of the proof determines the possible existence and arrangements of short chords. This is achieved by using a palette of reductions based on configurations of short chords. Then either the graph has only short chords or one can consider the shortest non-short chord, which we call a *medium* chord. The second half of the proof focusses on the segment of the outer cycle subtended by a medium chord, and finally we provide a reduction that shows one may assume there is no medium chord.

We will use several reductions applied to the graph G . There are two types of reductions. The first type of reduction is to remove a single edge to yield a 2-connected outerplanar graph G' . Then we appeal to the inductive hypothesis to produce an ID-set D' of G' of size at most $\frac{2}{5}n + \frac{1}{5}$ that we massage into an ID-set D of G such that the size of D is at most the size of D' .

The second type of reduction is to remove some vertices and add some edges to yield an outerplanar graph G' of order n' , which is 2-connected or K_2 . If $n' \geq 5$, then we again appeal to the inductive hypothesis to produce an ID-set D' of G' that we extend to an ID-set D of G . We say it is an (r, s) -reduction if D has r more vertices than D' and G has s more vertices than G' . If $r/s \leq \frac{2}{5}$, then the reduction is valid. The case where $n' \leq 4$ can be handled separately; alternatively, computer search shows that the bound of Theorem 1 is true for all graphs G of small order.

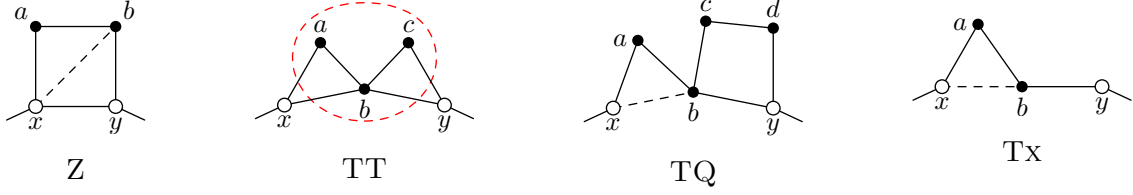


Figure 2: Some reducible configurations

3.2 External Triangles

In this subsection we consider reductions involving an external triangle: that is, a vertex of degree 2 whose neighbors are adjacent. The associated configurations are shown in Figure 2. In this picture a shaded vertex cannot have any other incident edge. We will show that one may assume that none of these configurations exists.

Lemma 3 *We may assume there is no portion $xaby$ of the outer cycle where xb and xy are chords. (Config. Z.)*

PROOF. Form graph G' by removing edge xb . Let D' be an ID-set of G' . Then either D' is an ID-set of G or becomes one on removing b . QED

Lemma 4 *We may assume there is no portion $xabcy$ of the outer cycle where xb and by are chords and b is incident to no other edge. (Config. TT.)*

PROOF. Form graph G' by removing $\{a, b, c\}$ and adding edge xy (if needed). If $|G'| \leq 4$ then the result is easily checked. Otherwise, let D' be an ID-set of G' . Without loss of generality assume $y \notin D'$. Then D' can be extended to an ID-set of G by adding b if $x \notin D'$, and by adding c if $x \in D'$. Thus this is a $(1, 3)$ -reduction. QED

Lemma 5 *We may assume there is no portion $abcdy$ of the outer cycle where xb and by are chords and b, c, d is incident to no other edge. (Config. TQ.)*

PROOF. Form graph G' by removing edge xb . Let D' be an ID-set of G' . This is an ID-set of G unless it contains both x and b . But then it must contain d and so $D - \{b\}$ is an ID-set of G . QED

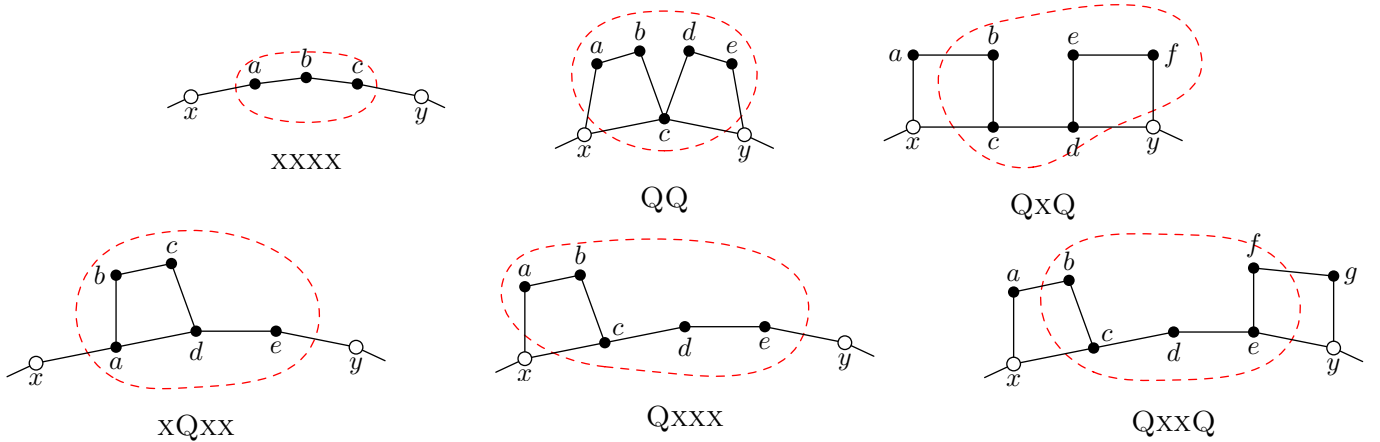


Figure 3: Some reducible configurations

Lemma 6 *We may assume there is no portion $xaby$ of the outer cycle where xb is a chord and b is incident to no other edge. (Config. Tx .)*

PROOF. Form graph G' by removing edge xb . Let D' be an ID-set of G' . This is an ID-set of G unless it contains both x and b . But then we can either just remove b or replace it with y and have an ID-set of G . QED

Thus we have shown that one may assume there is no chord joining vertices at distance two apart on the outer cycle.

3.3 External Quadrilaterals and Related Structures

In this subsection we consider reductions involving an external quadrilateral: that is, a 4-cycle containing two adjacent vertices of degree 2. The associated configurations are shown in Figure 3. In this picture a shaded vertex cannot have any other incident edges. We will show that one may assume that none of these configurations exists.

Lemma 7 *We may assume there is no portion $xabcy$ of the outer cycle where a, b, c have degree 2. (Config. $xxxx$.)*

PROOF. Form graph G' by removing $\{a, b, c\}$ and adding edge xy (if needed). If $|G'| \leq 4$ then the result is easily checked. Otherwise, let D' be an ID-set of G' . If

neither x nor y is in D' , then D' can be extended to an ID-set of G by adding b . If $x \in D'$ then add c , and if $y \in D'$ then add a . Thus this is a $(1, 3)$ -reduction. QED

Lemma 8 *We may assume there is no portion $abcdey$ of the outer cycle where xc and cy are chords and a, b, c, d, e are incident to no other edge. (Config. QQ .)*

PROOF. Form graph G' by removing $\{a, b, c, d, e\}$ and adding edge xy (if needed). If $|G'| \leq 4$ then the result is easily checked. Otherwise let D' be an ID-set of G' . Without loss of generality assume $y \notin D'$. Then D' can be extended to an ID-set of G by adding $\{b, e\}$. Thus this is a $(2, 5)$ -reduction. QED

Lemma 9 *We may assume there is no portion $abcdefy$ of the outer cycle where xc and dy are chords and a, b, c, d, e, f are incident to no other edge. (Config. QxQ .)*

PROOF. Form graph G' by removing $\{b, c, d, e, f\}$ and adding edge ay . If $|G'| \leq 4$ then the result is easily checked. Otherwise let D' be an ID-set of G' . Then D' can be extended by adding $\{d, f\}$ if $a \in D'$, and by adding $\{b, e\}$ if $a \notin D'$. Thus this is a $(2, 5)$ -reduction. QED

Lemma 10 *We may assume there is no portion $abcdey$ of the outer cycle where ad is a chord and a, b, c, d, e are incident to no other edge. (Config. $xQxx$.)*

PROOF. Form graph G' by removing $\{a, b, c, d, e\}$ and adding edge xy (if needed). If $|G'| \leq 4$ then the result is easily checked. Otherwise let D' be an ID-set of G' . Then D' can be extended by adding $\{b, e\}$ if $y \notin D'$, and by adding $\{a, c\}$ if $y \in D'$. Thus this is a $(2, 5)$ -reduction. QED

Lemma 11 *We may assume there is no portion $abcdey$ of the outer cycle where xc is a chord and a, b, c, d, e are incident to no other edge. (Config. $Qxxx$.)*

PROOF. Form graph G' by removing $\{a, b, c, d, e\}$ and adding edge xy (if needed). If $|G'| \leq 4$ then the result is easily checked. Otherwise let D' be an ID-set of G' . Then D' can be extended by adding $\{b, e\}$ if $y \notin D'$, and by adding $\{a, c\}$ if $y \in D'$. Thus this is a $(2, 5)$ -reduction. QED

Lemma 12 *We may assume there is no portion $xcdefgy$ of the outer cycle where xc and ey are chords and a, b, c, d, e, f, g are incident to no other edge. (Config. $QxxQ$.)*

PROOF. Form graph G' by removing $\{b, c, d, e, f\}$ and adding edge ag as well as edge xy (if needed). If $|G'| \leq 4$ then the result is easily checked. Otherwise let D' be an ID-set of G' . Without loss of generality assume $a \in D'$. It follows that $x \notin D'$. Then D' can be extended by adding $\{d, g\}$ if $y \notin D'$, and by adding $\{d, f\}$ if $y \in D'$. Thus this is a $(2, 5)$ -reduction. QED

3.4 Smooth Segments and Medium Chords

In subsection 3.2 we showed that there is no external triangle. Thus the only possibility for a short chord is one that joins vertices at distance 3 on the cycle; further, the intervening two vertices on the outer cycle cannot be incident with a chord (as it would create an external triangle) and thus they have degree 2. Define a *smooth segment* as a portion of the outer cycle such that (a) no interior vertex of the segment is incident with a non-short chord, and (b) every short chord incident with an interior vertex of the segment has its other end on the segment.

Lemma 13 *We may assume that every smooth segment has length at most 5. If the segment has length more than 3, then it contains exactly one short chord. Otherwise it contains at most one short chord.*

PROOF. Consider a smooth segment S . If it contains no short chord, then by Lemma 7 it has length at most 3. So assume S contains at least one short chord. By Lemma 8, two short chords on S cannot be incident. By Lemma 9, two short chords on S cannot be distance 1 apart. By Lemmas 10, 11 and 12, two short chords on S cannot be distance 2 or more apart. So it contains exactly one short chord. From the same lemmas it follows that S cannot have length 6 or more. QED

Lemma 14 *We may assume there is no portion $xabcy$ of the outer cycle where xy is a chord.*

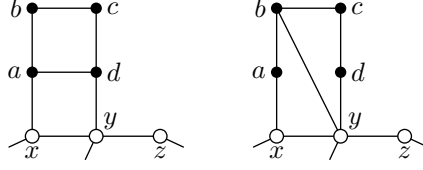


Figure 4: The two possibilities for a medium chord xy

PROOF. Form graph G' by removing $\{a, b, c\}$. If $|G'| \leq 4$ then the result is easily checked. Otherwise, let D' be an ID-set of G' . This can be extended to an ID-set of G by adding b if b is not dominated by D' , or by adding one of a or c as appropriate otherwise. Thus this is a $(1, 3)$ -reduction. QED

If every chord is short, then by (the proof of) Lemma 13 there can be at most one short chord and the outer cycle has at most seven vertices. And it is easily checked that one get the desired bound on $i(G)$ in this case.

So we may assume there is at least one non-short chord. Recall that a chord is medium if it is a shortest non-short chord. By the above two lemmas, we may assume that every medium chord xy spans a smooth segment of length 5 and contains exactly one short chord. The two possibilities up to symmetry for a medium chord xy are shown in Figure 4.

But we show next that in this case one can reduce.

Lemma 15 *We may assume there is no portion $abcdy$ of the outer cycle with chord xy and exactly one other chord that is either ad or by .*

PROOF. Let z be the vertex after y on the outer cycle. (If it does not exist then $n = 6$ and the result is true.) Form graph G' by removing $\{a, b, c, d, y\}$ and adding edge xz (if needed). If $|G'| \leq 4$ then the result is easily checked. Otherwise let D' be the small ID-set of G' . If $x \in D'$, then D' can be extended by adding c and, if not already dominated, adding z . If $x \notin D'$ and y is dominated by D' , then D' can be extended by adding a and c . If y is not dominated by D' , then D' can be extended by adding $\{b, y\}$ if the chord is ad , and by adding $\{a, d\}$ if the chord is by . Thus this is a $(2, 5)$ -reduction. QED

And thus we have reached our conclusion. This completes the proof of the bound in Theorem 1.

3.5 Extremal Graphs

To show that the bound in Theorem 1 is best possible, consider the graph W_r defined as follows for $r \geq 1$. Start with a star centered at vertex u with neighbors $v_1, \dots, v_{r+1}$. Then for each $i \in \{1, \dots, r\}$, introduce a quadrilateral and add two edges joining v_i and v_{i+1} to adjacent vertices on the quadrilateral. For example, the graph W_4 is shown in Figure 5.

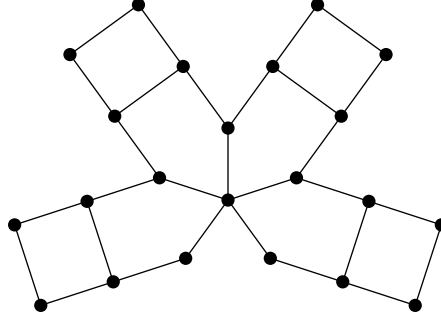


Figure 5: The outerplanar graph W_4

Lemma 16 *The graph W_r is 2-connected outerplanar and has $5r + 2$ vertices. Further, $i(W_r) = 2r + 1$.*

PROOF. Consider any ID-set D of W_r . If D contains vertex u , then it must also contain two vertices from each quadrilateral, for a total of $2r+1$ vertices. If D does not contain vertex u , then it must also contain one degree-two vertex from each quadrilateral; further, the v_i have disjoint closed neighborhoods (excluding u), and so there are at least another $r + 1$ vertices in D . QED

It is unclear what all the extremal graphs are. By computer search there are two others for $n = 7$ and one other for $n = 12$, shown in Figure 6.

4 Minimum Degree Two

We conclude with some thoughts on another family of outerplanar graphs, namely the set of all outerplanar graphs of minimum degree at least 2. Now, there are both sparse and dense cases with small independent domination, as we noted

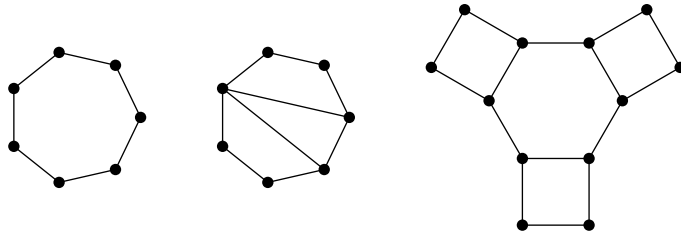


Figure 6: Some outerplanar graphs with maximum i

earlier that maximal outerplanar graphs G have $i(G) \leq n/3$, while the cycle C_n has $i(C_n) = \lceil n/3 \rceil$. The best bound we currently know (though surely it is not best possible) is $i(G) \leq n/2$. Indeed, in [6] we showed that such a graph has two disjoint independent dominating sets.

The largest value of the independent domination number that we know of is asymptotically $\frac{4}{9}n$. Consider the following graph Q_m . Take a path $v_1 v_2 \dots v_m$. For each i between 1 and $m - 1$, add vertices a_i and a'_i and use them in a path of length 3 joining v_i and v_{i+1} , and add vertices b_i and b'_i and similarly use them in a path of length 3 joining v_i and v_{i+1} . Then for every vertex v_i , introduce a 4-cycle C and join by one edge to v_i . The graph Q_6 is shown in Figure 7.

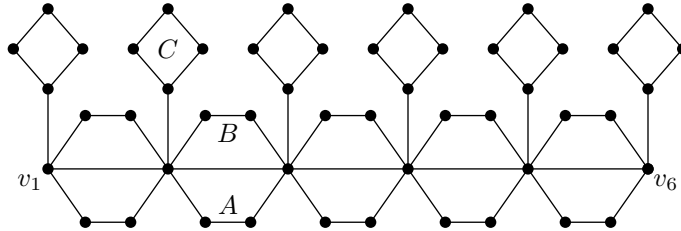


Figure 7: An outerplanar graph Q_6 with independent domination about $\frac{4}{9}$ its order

Lemma 17 *The graph Q_m has order $n = 9m - 4$ and $i(Q_m) = (4n - 2)/9$.*

PROOF. In order to dominate C_i , a dominating set must contain at least two vertices from the closed neighborhood $N[C_i]$. Further, any independent dominating set of Q_m must contain exactly one vertex of every $a_i a'_i$ pair and of every $b_i b'_i$ pair. Thus $i(G) \geq 2m + 2(m - 1) = 4m - 2$. It is easily to check that there exists an ID-set of size $4m - 2$. QED

We note that both the above example and the extremal graphs for Theorem 1 contain many induced 4-cycles. Thus it might be interesting to consider the case where such are excluded.

References

- [1] C.N. Campos and Y. Wakabayashi, On dominating sets of maximal outerplanar graphs. *Discrete Appl. Math.* 161 (2013), 330–335.
- [2] M. Dorfling, J.H. Hattingh, and E. Jonck, Total domination in maximal outerplanar graphs II. *Discrete Math.* 339 (2016), 1180–1188.
- [3] M. Dorfling, J.H. Hattingh, and E. Jonck, Total domination in maximal outerplanar graphs. *Discrete Appl. Math.* 217 (2017), 506–511.
- [4] S. Fisk, A short proof of Chvátal’s watchman theorem. *J. Combin. Theory Ser. B* 24 (1978), 374.
- [5] W. Goddard and M.A. Henning, Independent domination in graphs: A survey and recent results. *Discrete Math.* 313 (2013), 839–854.
- [6] W. Goddard and M.A. Henning, Independent domination, colorings and the fractional idomatic number of a graph. *Appl. Math. Comput.* 382 (2020), 125340.
- [7] S.M. Hedetniemi, S.T. Hedetniemi and D.F. Rall, Acyclic domination, *Discrete Math.* 222 (2000), 151–165.
- [8] C. Liu, A note on domination number in maximal outerplanar graphs. *Discrete Appl. Math.* 293 (2021), 90–94.
- [9] G. MacGillivray and K. Seyffarth, Bounds for the independent domination number of graphs and planar graphs, *J. Combin. Math. Combin. Comput.* 49 (2004), 33–55.
- [10] M.D. Plummer, D. Ye, and X. Zha, Dominating maximal outerplane graphs and Hamiltonian plane triangulations. *Discrete Appl. Math.* 282 (2020), 162–167.
- [11] S. Tokunaga, Dominating sets of maximal outerplanar graphs. *Discrete Appl. Math.* 161 (2013), 3097–3099.